

ROBOT PART ITERATION LOG - GRADING RUBRIC

Total Points: 20

HEADER INFORMATION & BASIC DOCUMENTATION (2 points)

Points	Criteria
2	All fields complete: Name, Group, Part Name, Version #, Manufacturing Technique selected
1	Missing 1-2 fields
0	Missing 3+ fields

1. WHAT DID YOU CHANGE? (4 points)

Points	Criteria
4	Clear sketch OR 2+ bullet points showing specific changes; easy to understand what was modified from previous version
3	Adequate documentation but lacks clarity OR only 1 bullet/minimal sketch
1-2	Vague description (e.g., "made it better") without specific details
0	Section blank or completely unclear

2. CALIPER MEASUREMENTS (6 points)

2 points per measurement row

Points	Criteria
2	Spot clearly described, Target value given, Actual measurement recorded, Error calculated correctly ($\pm$ value), Initials present
1	Missing 1-2 elements (e.g., no error calculation OR no initials OR unclear spot description)
0	Blank OR missing 3+ elements

Note: All 3 measurement rows must be completed for full section credit.

3. FIT / FUNCTION TEST (4 points)

2 points per feature/quality identified and evaluated

Points	Criteria
4	Two distinct features/qualities identified with clear, specific evaluation criteria (e.g., 'Motor shaft fits snugly with no play' or 'Deck clears battery by 5mm minimum')
3	Two features identified but one lacks specificity OR evaluation criteria are vague
2	Two features identified but both lack clear evaluation criteria OR only one feature properly documented
1	Minimal effort (generic statements like 'it fits' without any detail)

0	Section blank or no meaningful evaluation
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4. OBSERVATION + NEXT MOVE (4 points)

Points	Criteria
4	Both sentences complete and thoughtful; shows genuine reflection on what surprised them AND specific plan for next version
3	Both sentences present but one lacks depth (generic response like "nothing surprised me")
2	Only one sentence completed OR both are very brief/superficial
1	Minimal effort (fragments or unclear responses)
0	Section blank

GRADING NOTES:

- **Professionalism matters:** Illegible handwriting or sloppy work may result in up to -1 point deduction
- **Version tracking:** Students should be completing this for EACH iteration of their part (typically 2 worksheets per student as noted in assignment)
- **Collaboration evidence:** While the form has no signature line, measurements should show evidence of peer verification (initials column)
- **Fit/Function features:** Look for specific, measurable criteria. Good examples: 'Bolt holes align within 0.5mm', 'Part clears obstacle by 3mm', 'No wobble when motor spins'. Poor examples: 'looks good', 'works fine'

Point Scale:

Point Range	Performance Level
18-20	Excellent documentation
15-17	Good, meets expectations
12-14	Adequate, needs improvement
Below 12	Incomplete or insufficient effort